IDNet: Intensity-Constrained Detail-Enhanced Network for Hyperspectral and LiDAR Collaborative Classification

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***Abstract*— The complementarity of hyperspectral image (HSI) and light detection and ranging (LiDAR) data provides significant advantages in land cover classification tasks. Currently, most HSI and LiDAR classification algorithms focus on local–global feature learning frameworks. However, high-frequency characteristics play a vital role in distinguishing different land cover classes. For instance, the high-frequency information in the spectral curve correspond to spectral features characterized by rapid variations, as well as detailed features (edges and boundaries) in the spatial land cover information. Therefore, embedding high-frequency detail information can enhance the collaborative classification of HSI and LiDAR data. Motivated by this, this article proposes an intensity-constrained detail-enhanced network, named IDNet. This network embeds Laplacian high-frequency features into a Transformer network architecture and designs Transformer blocks with dynamic range awareness and similarity-based spatial–spectral feature grouping interactions to efficiently learn local–global high-frequency spatial–spectral features in the HSI and LiDAR branches. During the feature fusion stage, differential convolution is utilized to construct an enhanced detail expert that emphasizes high-frequency feature information, enabling adaptive dynamic feature fusion to improve the discriminability of the fused features. Experimental results show that compared with the current state-of-the-art algorithms, such as S2ENet, HCT-Net, and MHST, the proposed IDNet is improved by 0.55%–2.72%, 0.17%–1.31%, 0.41%–4.92%, and 2.58%–13.13% on the Houston 2013, Trento, MUUFL, and Augsburg datasets with OA as the evaluation index, respectively. Code of this project is at: https://github.com/ZhaoYuQing01/IDNet**

***Index Terms*— Classification, hyperspectral image (HSI), light detection and ranging (LiDAR), mixture of experts (MoE), Transformer.**

I. Introduction

HYPERSPECTRAL and light detection and ranging (LiDAR) data are heterogeneous data types with different imaging mechanisms [1], [2], [3]. Hyperspectral image (HSI) is characterized by high spectral resolution, allowing for a detailed representation of the spectral reflectance properties of surface features [4], [5]. In contrast, LiDAR data could capture the 3-D structural information of land cover features [6], [7]. Currently, leveraging the complementarity of these two data types, i.e., utilizing height information to assist in the identification of spectrally similar objects or applying spectral information to identify land covers with similar heights, has become one of the key research hotspots.

Due to the complexity of land-cover scenes and the influence of environmental factors, achieving collaborative classification with HSI and LiDAR still faces significant challenges.

1) The spectral and spatial features of HSI and LiDAR have entirely different physical meanings, making the design of efficient feature learning strategies a key issue.
2) How to effectively utilize the features of these two data types to construct complementary multimodal fusion features will significantly impact the effectiveness of land cover classification.

Due to the rapid development of deep learning, convolutional neural networks (CNNs) have shown great potential in joint hyperspectral and LiDAR classification, thanks to their powerful local feature extraction capabilities. Xu et al. [8] introduced a dynamic multimodal gradient optimization strategy. In the multimodal feature fusion, Zhang et al. [9] designed and cascaded a novel interleaving perception convolutional layer to integrate HSI and LiDAR statistics.

Although spatial–spectral feature learning strategies based on CNNs can improve the classification performance of HSI and LiDAR data, the global feature modeling is indispensable for characterizing land cover features. Therefore, Transformers, due to their powerful global feature modeling capability, could provide a more comprehensive characterization of land cover properties. Numerous studies have concentrated on the spectral–spatial feature learning of HSI and LiDAR [10], [11], with the CNNs–Transformer framework demonstrating remarkable effectiveness in the collaborative classification of HSI and LiDAR data. This is primarily because the local features extracted by CNNs and the global contextual information captured by the Transformer complement each other, thereby enhancing the model's ability to model complex patterns and long-range dependencies. CNNs could first capture low-level features from HSI or LiDAR data, and

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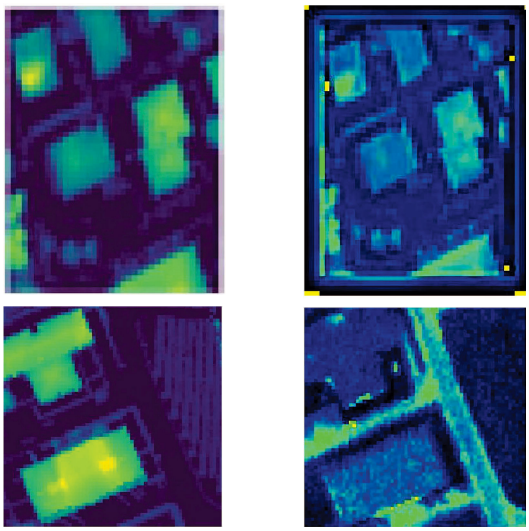


Fig. 1. High-frequency features of HSI patches. The areas in the graph where the color changes abruptly and the gray value changes sharply are where the high-frequency components are located. The edge of the blue and yellow junctions is where the high-frequency components are concentrated, such as the boundary area of the cube shape.

then, the Transformer further enhances the learning of global information, allowing the model to better handle the complex structures in HSI and LiDAR data [12], [13], [14]. Wang et al. [15] focused on local–global feature learning, embedding multiscale asymmetric depthwise convolutional layers and innovative local–global Transformer networks for describing HSI and LiDAR land covers. Song et al. [16] proposed a hierarchical feature fusion and separation blocks followed by a dual-structure feature encoder structure (CNNs-based spatial encoder and Transformer-based spectral encoder) guided by height information of LiDAR.

HSI and LiDAR data typically contain rich spatial and spectral information, and fixed convolutions cannot adapt flexibly to complex landform scenarios [12], [17]. Therefore, by applying the concept of dynamic parameters, a convolutional mechanism can be designed to precisely extract local features, enhancing the model's generalization ability across different scenes. In addition, the HSI data often have high dimensionality, with many bands containing redundant information. Through a similarity-based feature grouping mechanism, these redundant features can be effectively aggregated, thus reducing the feature dimensionality and computational complexity, while retaining the most meaningful information. Different land covers may exhibit similar spatial features in the LiDAR data, and similarity-based feature grouping interactions can help classification models distinguish these subtle differences.

The Laplacian operator highlighting the high-frequency components in the data, particularly in sharp regions, as illustrated in Fig. 1. It is useful for identifying fine variations and details within the data, making it commonly used for extracting detailed information or analyzing the local structure of the data. Motivated by this and a dynamic similarity-based feature grouping mechanism, this article focuses on the high-frequency detail feature learning in the spatial–spectral domain and proposes an intensity-constrained detail-enhanced network, named IDNet. Single-modal LiDAR remote sensing data effectively exploits spatial information, but the inherent absence of spectral features fundamentally restricts the further improvement of land cover classification accuracy. In contrast, IDNet overcomes this limitation by innovatively integrating spatial–spectral features that incorporate depth information. By taking into account the spatial structure information and spectral reflection characteristics, the network not only breaks through the limitations of a single data source but also enhances the ability to capture high-frequency details under the constraint of intensity, thereby significantly improving the accuracy and reliability of land cover classification and providing a new method for the multimodal fusion of remote sensing data and the fine identification of complex land cover types [18]. The main contributions of this article are threefold.

1) We embed Laplacian high-frequency features into the Transformer network architecture and design a Transformer block with dynamic range perception and similarity-based spatial–spectral feature grouping interactions in hyperspectral feature learning.
2) We construct a spatial feature grouping interaction Transformer block that effectively models high-frequency LiDAR spatial features and dynamically perceives effective regions.
3) We utilize differential convolutions and a dynamic selection mechanism to construct enhanced detail experts that emphasize high-frequency feature statistics of the fused features.

This article is organized as follows. Section II introduces the proposed IDNet, Section III presents the specific experimental results and analysis, and Section IV concludes this article.

II. Related Works

A. Dual-Branch Architectures Based on CNNs

Dual-branch architectures based on CNNs are commonly used in HSI and LiDAR classification tasks [19]. The convergence imbalance in dual-branch features can result in uneven feature learning. To address this issue, the dual-branch convolutional feature learning framework helps the proposed model achieve superior classification performance. Based on the morphological feature learning, a 3-D convolutional layer and morphological-related block [20] were employed to capture the spatial–spectral characteristics by considering the elevation feature of LiDAR. Moreover, Hong et al. [21] focused on the encoder–decoder network architecture and integrated multimodal information by requiring the fused features to sequentially reconstruct the original multimodal inputs. Moreover, three deep feature fusion strategies were verified. Gao et al. [22] also designed various stages of HSI and LiDAR feature learning, and they utilized a dual-branch convolutional feature learning structure based on dual-branch feature extraction, multiscale hierarchical fusion, and an adaptive feature fusion module. Especially, the attention mechanism with convolutional layers plays an important role in the multibranch feature learning and fusion, related works include, FusAtNet [23], multiattentive hierarchical fusion network (MAHiDFNet) [24], multimodal attention-aware CNNs [25],

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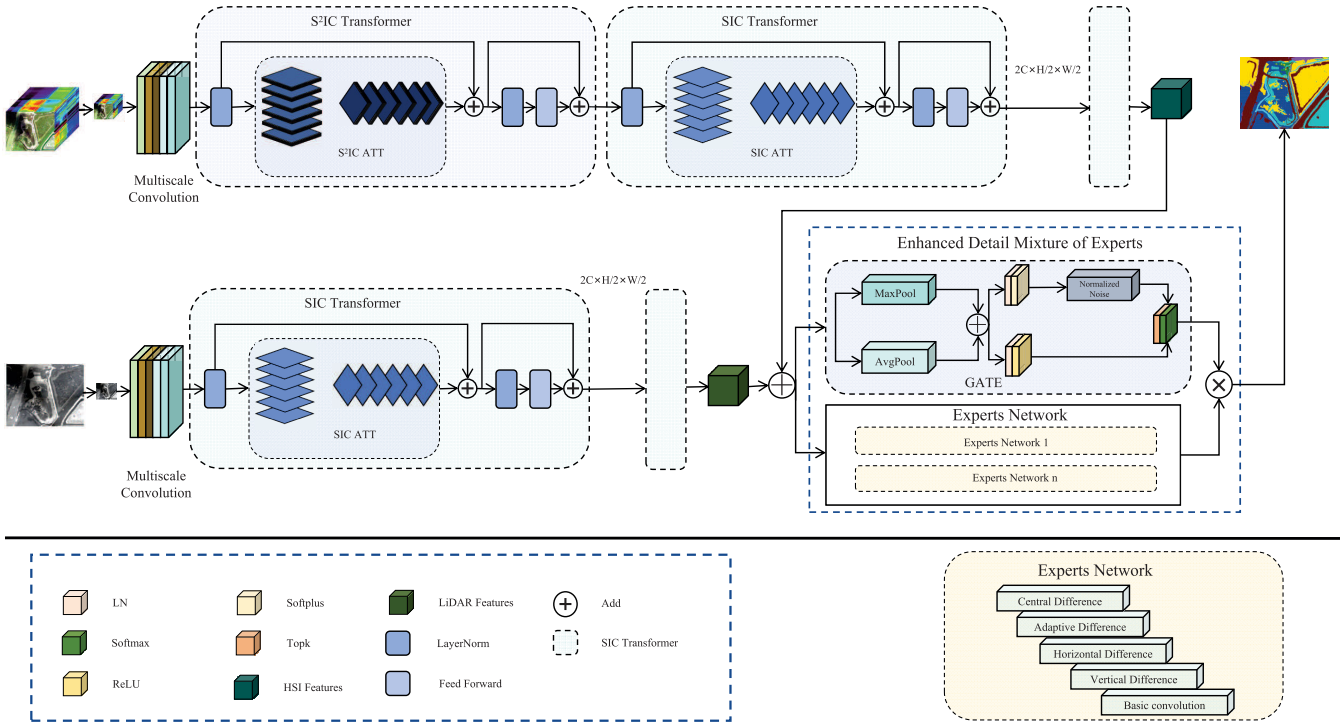


Fig. 2. Architecture of IDNet, including multiscale convolution, S^{2}IC block, SIC block, and EDME module.

MSLAENet [26], mutual guidance attention (MGA) module [27], bilinear attention fusion [28], and attention-guided fusion and classification [29]. In addition, some studies embed other feature learning approaches into CNNs. For instance, random walk [30], hashing-based deep metric learning [31], and graph convolutional networks [32], and Lu et al. [33] incorporated a dual-branch coupled adversarial learning mechanism into CNNs to extract more compact and discriminative features of HSI and LiDAR data.

B. Dual-Branch Architectures Followed by CNNs–Transformer Feature Learning Style

Dual-branch architectures adopting the CNNs–Transformer feature learning style first utilized CNNs to extract local features and then embed Transformer structures to capture global features and perform feature fusion. Tang et al. [34] proposed a CNNs-cross-modal attention fusion Transformer network with a semantic-guided feature fusion method for merging deep features from HSI and LiDAR branches. Moreover, for considering the modality-specific and modality-shared features of HSI and LiDAR data, Sun et al. [35] mainly designed a convolution–Transformer module, including a morphological feature enhancement block and dilated convolutional layer, for efficient feature interaction. Simultaneously, a series of lightweight operations, i.e., grouped operations, are embedded into the proposed network. For feature interaction, the interactive enhanced network [36] incorporated graph convolution and multihead self-attention (MSA) to enhance the collaborative feature learning between LiDAR and HSI. Other CNNs–Transformer networks, such as, binary-tree Transformer network [37], multiscale head selection Transformer [38], cross-modal enhanced CNNs and Transformer [39], frequency-domain-based network [40], multimodal informative ViT [41], local information interaction Transformer [42], and diffusion-enhanced CNNs–Transformer [43], have been widely explored in recent studies.

Combining CNNs and Transformer networks could fully leverage the strengths of both in feature extraction and long-range dependency modeling, significantly improving the classification accuracy in HSI and LiDAR data classification. However, most networks approach the problem from the perspective of spatial–spectral feature learning, neglecting the high-frequency detail information of both datasets. In HSI and LiDAR collaborative classification, capturing and combining high-frequency information helps improving the overall classification performance.

III. Methodology

As illustrated in Fig. 2, IDNet is specifically designed for the joint classification of hyperspectral imagery (with multiple spectral channels) and LiDAR point cloud data [represented as digital surface models (DSMs)], of absolute heights above sea level). The model is primarily composed of three key components: S^{2}IC block, SIC block, and EDME module. Both the SIC and S^{2}IC modules incorporate novel attention mechanisms that significantly enhance the representation of fine details. These mechanisms leverage the Laplacian pyramid and dynamic convolution to efficiently capture contextual information, resulting in a highly optimized and robust Transformer module. The EDME module further refines feature extraction by applying differential convolution to enhance

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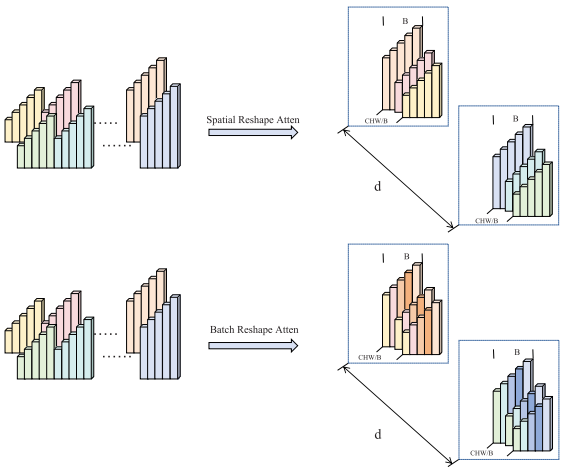


Fig. 3. Structure of the two reshaped attention mechanism.

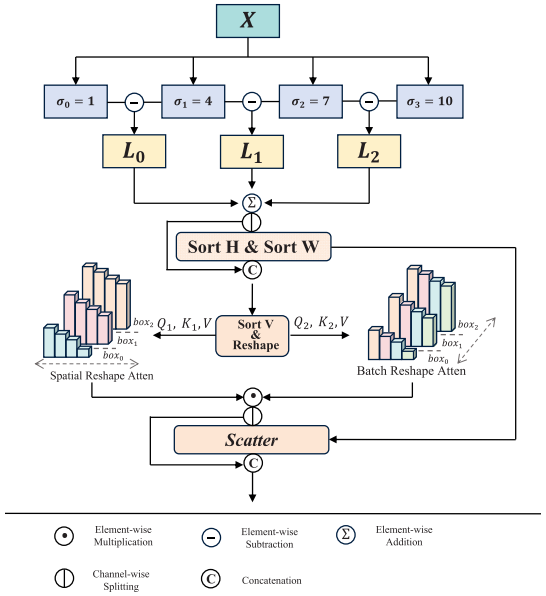


Fig. 4. Architecture of spatial intensity-constrained attention mechanism. Spatial–spectral intensity-constrained block is similar in that it has an additional spectral channel dimension.

high-frequency feature information. In addition, a dynamic selection mechanism facilitates adaptive and effective feature fusion.

A. Data Preprocessing

The hyperspectral data often contains tens-to-hundreds of bands, which increases computational complexity, especially when hyperspectral data and LiDAR data need to be jointly classified and processed simultaneously. Dimensionality reduction is a common preprocessing step for hyperspectral data. Principal component analysis (PCA) is widely used here: it converts original bands into uncorrelated principal components via linear transformation. This process reduces data redundancy while preserving the main spectral features. PCA achieves data compactness and noise reduction by projecting original features onto principal components derived from

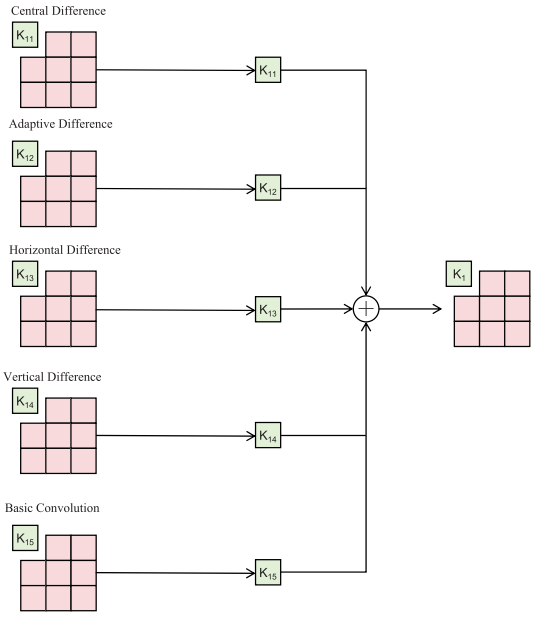


Fig. 5. Details of the expert network. The weights and biases of the five convolutions are added together to form the final convolutional weights and biases.

eigenvalue decomposition. Specifically, it transforms high-dimensional data points (i.e., samples with spectral features) into a new coordinate system where each principal component corresponds to a direction of maximum variance, weighted by eigenvalues. At the same time, the dimensionality of hyperspectral data after dimensionality reduction is reduced, making it easier to align and fuse the features with LiDAR data.

B. Spatial–Spectral Intensity-Constrained Block

In hyperspectral feature learning, the conventional self-attention mechanism in Transformers typically employs a fixed attention range, limiting its adaptability in capturing long-range dependencies among related features. In addition, its inherent low-pass filtering behavior suppresses high-frequency information, thereby constraining feature expressiveness and reducing classification accuracy, particularly in complex scenarios. In the HSI and LiDAR images joint classification task, where the data exhibit different spectral characteristics and varying extraction ranges, we have explored innovative approaches aimed at overcoming these limitations, emphasizing the preservation of high-frequency information and the expansion of the attention range.

We have innovatively developed the spatial–spectral intensity-constrained block, which seamlessly integrates Laplace high-frequency features into the advanced architecture of the Transformer network. Moreover, we have designed a unique Transformer block that not only enables dynamic range perception but also facilitates similarity-based spatial–spectral feature grouping and interaction. By leveraging the inherent complementarity of global features, this Transformer block efficiently learns and extracts intricate local high-frequency

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spatial–spectral features, significantly enhancing the overall performance of the hyperspectral feature analysis.

In the multimodal image fusion, the pyramid segmentation attention module has proven effective in capturing high-frequency imagery details, significantly enhancing the quality of the fused features. Since the traditional Transformer model primarily emphasizes low-frequency information, we propose the construction of a Laplace pyramid to capture different frequency levels within the feature map [44]. This process involves extracting normalized features based on the variance values of Gaussian functions at varying scales. The outputs from layer i and layer $i+1$ are then subtracted to acquire the Laplace pyramid, with each level representing encoded information at different scales

$$\boldsymbol{G}_k = \text{GaussianBlur}(\boldsymbol{X}, \sigma_k) \tag{1}$$

$$\text{GaussianBlur}(\boldsymbol{X}, \sigma_k) = \boldsymbol{X} * \frac{1}{\sqrt{2\pi\sigma_k^2}} \mathrm{e}^{-\frac{x^2+y^2}{2\sigma_k^2}} \tag{2}$$

$$\boldsymbol{G}_{k+1} = \text{Downsample}(\boldsymbol{G}_k) \tag{3}$$

$$\boldsymbol{L}_k = \boldsymbol{G}_k - \text{Upsampling}(\boldsymbol{G}_{k+1}) \tag{4}$$

where $\boldsymbol{X}$ represents the input feature map, x and y correspond to the spatial position in the encoded feature map, sigma_k is the variance of the Gaussian function of the kth scale, and the symbol $*$ represents the convolution operator. $\boldsymbol{L}_k$ and $\boldsymbol{G}_k$ denote the kth level in the pyramid of Laplace and the kth layer in Gauss's pyramid, respectively. $\boldsymbol{G}_{k+1}$ is the $k+1$ level in the Gaussian pyramid. "Upsampling" refers to an operation that increases the spatial resolution of the feature map to match that of $\boldsymbol{G}_k$.

Through the Laplace pyramid, Gaussian blur, and differential operations can be used to separate high-frequency information, such as edges in the image, which improves the shortcomings of traditional pixel-based global or local features that only focus on low-frequency information. In addition, the multilevel pyramid structure is used to capture the spatial structure features of different scales, making up for the vulnerability to noise interference and limited detail expression ability.

To achieve self-attention across an adaptive long range, we incorporate dynamic convolution and employ a dual-path processing strategy. Before applying the traditional convolutional operation, the input features are rearranged along the channel dimension and split into two branches. The first branch undergoes both horizontal and vertical sorting as illustrated in Fig. 3, and the results are then merged with the second branch to produce the reorganized features. A separable convolutional operation is then applied, arranging high- and low-intensity pixels into regular patterns at the diagonal corners of the matrix, allowing the convolution to effectively acquire across the dynamic range.

The spatial–spectral intensity-constrained Transformer module incorporates a comprehensive processing approach that integrates spatial and spectral information tailored to the unique characteristics of hyperspectral images. The input data undergo meticulous processing through sorting, grouping, reshaping, and attention calculation steps, thereby fully leveraging the spatial and spectral attributes of the image. Initially, the first half of the input is sorted along the dimensions of spectral channel, height, and width, with the sorting indices being recorded. Subsequently, depth-separable convolution is applied to compute the query ($\boldsymbol{Q}$), key ($\boldsymbol{K}$), and value ($\boldsymbol{V}$). The resultant tensors are divided into five components: $\boldsymbol{Q}_1$, $\boldsymbol{K}_1$, $\boldsymbol{Q}_2$, $\boldsymbol{K}_2$, and $\boldsymbol{V}$. Among these, $\boldsymbol{Q}_1$, $\boldsymbol{K}_1$, $\boldsymbol{Q}_2$, and $\boldsymbol{K}_2$ are utilized for computing two different attention mechanisms, while $\boldsymbol{V}$ is employed to generate the final output

$$\boldsymbol{V}, \text{id} = \text{Sort}\left(\boldsymbol{V}'\right) \tag{5}$$

$$\text{Attention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \text{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{d_k}}\right)\boldsymbol{V} \tag{6}$$

$$\boldsymbol{R}_i = \text{Gather}\left(\boldsymbol{R}_i', i\ d\right) \tag{7}$$

where $\boldsymbol{V}'$ and $\boldsymbol{R}_i'$ represent the reshaping feature operation, id denotes the index of the sorted value, and $\boldsymbol{R}_i$ stands for $\boldsymbol{K}_1$, $\boldsymbol{K}_2$, $\boldsymbol{Q}_1$, and $\boldsymbol{Q}_2$. Gather is the operation of collecting elements from the tensor in a specified dimension based on the given index. $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$ are of the shape (B, num_heads, d $\times$ factor, and $C\times$ $H\times$ W/factor).

The value tensor ($\boldsymbol{V}$) is sorted based on the previously recorded indices, and these indices are also utilized to rearrange the query and key tensors. Attention calculation is then performed on these two sets of queries and keys. Following this, the sorted output is mapped back to its original position using the indices. The two outputs are multiplied, and a part of the input is substituted back into the first half of the original input channel, thereby completing the whole forward propagation.

In the attention calculation section, the shapes of the query ($\boldsymbol{Q}$), key ($\boldsymbol{K}$), and value ($\boldsymbol{V}$) tensors are reorganized, and attention weights are computed. The tensors are padded, and the product of the adjusted number of channels, height, and width is calculated. The tensors are rearranged in two ways, and the value tensor ($\boldsymbol{V}$) is weighted and summed based on the attention weights. Padding is removed to obtain the processed output tensor.

C. Spatial Intensity-Constrained Block

The spatial intensity-constrained Transformer module shares similarities with the spatial–spectral intensity-constrained Transformer module as depicted in Fig. 4. The spatial intensity-constrained Transformer module exhibits similarities to the spatial–spectral intensity-constrained Transformer module in that both are designed to leverage and enhance specific aspects of data processing. While the spatial–spectral intensity-constrained Transformer module primarily focuses on integrating spatial and spectral features, with an emphasis on the spatial characteristics of high-frequency hyperspectral data, the spatial intensity-constrained Transformer module takes a more streamlined approach. It specifically targets the spatial features of high-frequency LiDAR, aiming to improve the extraction and utilization of global features within these spatial dimensions. By incorporating intensity constraints within its Transformer architecture, this module can refine and emphasize the most significant spatial patterns and features, thereby enhancing the high-frequency feature learning of LiDAR data.

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Subsequently, the output of the dynamic convolution is divided into value features and two sets of query keys, which are then reshaped into tensors of dimensions $C \times B \times (HW/B)$. To capture both global and local information, we adopt two reshaping methods. The first method classifies spatial elements into boxes, each containing the same number of elements. By setting the number of boxes to B, each box encompasses HW/B elements, allowing for the extraction of large-scale information. Within each box, a significant number of dynamically positioned pixels contribute to capturing broader contextual features. The second method sets the frequency of each box to B, resulting in HW/B boxes. This configuration captures fine-grained information, with each box containing a small number of adjacent intensity pixels. The outputs of the two sets of query key features are then multiplied by the two types of reshaped tensors, followed by a self-attention mechanism, to produce the final output

$$X_1 = \text{Softmax}\left(\frac{\boldsymbol{Q}_1'\left(\boldsymbol{K}_1'\right)^T}{\sqrt{n}}\right)\boldsymbol{V}' \tag{8}$$

$$X_2 = \text{Softmax}\left(\frac{\boldsymbol{Q}_2'\left(\boldsymbol{K}_2'\right)^T}{\sqrt{n}}\right)\boldsymbol{V}' \tag{9}$$

$$X = X_1 * X_2 \tag{10}$$

where n represents the number of heads, $\boldsymbol{K}_1'$, $\boldsymbol{K}_2'$, $\boldsymbol{Q}_1'$, $\boldsymbol{Q}_2'$, and $\boldsymbol{V}'$ denote the results of the reshaping operation corresponding to $\boldsymbol{K}_1$, $\boldsymbol{K}_2$, $\boldsymbol{Q}_1$, $\boldsymbol{Q}_2$, and $\boldsymbol{V}$, respectively. $*$ stands for multiplication by element operation.

D. Enhanced Detail Mixture of Experts (MoE)

To enhance the flexibility, scalability, and generalization capability of the proposed model, while optimizing multimodal feature fusion, improving computational efficiency and performance, and strengthening robustness, we adopt a hybrid expert system architecture known as MoE to generate the final classification results. This architecture leverages many frequency learning experts to integrate input features and adapt to diverse content within remote sensing images. The gating network (GATE) assigns varying weights for feature fusion and selects the most suitable expert based on the input features [45]. To prevent homogenization among experts, the gating system produces sparse weight distributions.

The GATE system is given as follows. In the enhanced detail MoE (ED-MoE) framework, global average pooling (GAP), and global max pooling (GMP) jointly enable the multiscale feature fusion: GAP aggregates spatial information for global context, whereas GMP highlights discriminative local features, ensuring synergistic feature representation. In addition, the gated network can dynamically select experts to process information of different granularities according to different characteristics of GAP and GMP. Therefore, it first applies GAP and global maximum pooling to the input features. The resulting pooled features are summed, passed through a learned matrix, and activated using the SoftPlus and ReLU functions to alleviate the problem of vanishing gradients and improve the efficiency of the model. Next, the features are achieved by a fully connected layer. We employ learnable noise to achieve approximately uniform selection probabilities for each expert. The outputs are then summed, and the top-k selection method is employed to identify the n experts with the highest weight values among the k positions, assigning negative infinity to the remaining positions. Finally, softmax is applied to compute the expert weights, with the weights of unselected experts set to 0

$$\boldsymbol{Y}_1 = \text{AAP}(\boldsymbol{Y}) + \text{AMP}(\boldsymbol{Y}) \tag{11}$$

$$\boldsymbol{Y}_2 = \text{SoftPlus}(\text{LN}(\boldsymbol{Y}_1)) + \text{ReLU}(\text{LN}(\boldsymbol{Y}_1)) \tag{12}$$

$$W_k = \text{SoftMax}\ (\text{Topk}(Y_2)) \tag{13}$$

where $\text{AAP}(\cdot)$ and $\text{AMP}(\cdot)$ correspond to the adaptive average pooling and adaptive maximum pooling operations, respectively, and $\boldsymbol{Y}_1$ is obtained by summing. The normalized noise obtained by operations, such as SoftPlus and $\boldsymbol{Y}_1$ through the fully connected layer is combined, and the top-k operation is applied. Finally, the SoftMax activation function sets the weight of the unselected expert to 0.

To capture high-frequency information, we designed a detail-enhancement convolutional operation for the expert system as illustrated in Fig. 5. This operation integrates five parallel convolutions: four differential convolutions and one standard convolution for feature extraction. The standard convolution captures intensity-level information, while the differential convolutions enhance gradient-level information to augment the representation and generalization capabilities of the standard convolution. Specifically, we utilize central difference convolution and angular difference convolution, implemented by rearranging the learned kernel weights. Moreover, horizontal and vertical differential convolutions perform differential operations along these respective axes, explicitly encoding gradient information into the convolutional layer. By learning advantageous gradient features, these differential convolutions enhance the directional feature extraction. The learned kernel weights are subsequently rearranged, and the convolutions are directly applied to the original input features

$$F_k = Y * \sum_{i=1}^{5} K_{\text{ji}},\ j = 1, 2, \ldots, 9 \tag{14}$$

$$Z = \text{Conv}\left(\sum_{k=1}^{n} W_k \cdot F_k\right) \tag{15}$$

where $\sum_{i=1}^{5}$ represents the basic convolution, central difference convolution, adaptive difference convolution, horizontal difference convolution, and vertical difference convolution converted kernels, respectively, and then, the parallel convolutions are combined, and $*$ represents the convolutional operation. F_i represents the ith experts network. Then, this expert network could capture the detailed statistics of HSI and LiDAR fusion features.

IV. Experiments

A. Dataset

In this experiment, four commonly used HSI and LiDAR datasets are utilized to evaluate the effectiveness of the proposed IDNet. The detailed information of the data are shown in Table I.

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TABLE I
DATASET INFORMATION

Category	Information Type/Characteristic	Houston2013	Trento	MUUFL	Augsburg
HSI	Sensor Name	ITRES CASI-1500	Eagle sensor	ITRES CASI-1500	DAS-EOC
	Spatial Resolution	2.5m	1m	0.54×1.0m	30m
	Wavelength Range	0.38-1.05μm	0.42-0.99μm	0.38-1.05μm	0.4-2.5μm
	Number of Spectral Bands	144	63	64	180
LiDAR	Sensor Name	ITRES CASI-1500	Optech ALTM 3100EA	ALTM	DLR-3 K
	Spatial Resolution of DSM	2.5m	1m	0.60×0.78m	30m
Scene Classification	Scene Size(Pixels)	349×1905	166×600	325×220	332×485
	Location	University of Houston, USA	Rural areas in southern Trento, Italy	University of Southern Mississippi, USA	Augsburg, Germany
	Number of Classes	15	6	11	7
	Ground Truth Samples	15029	30214	53687	78294

1) *Houston2013 Dataset:* The Houston2013 dataset [46], [47], which captured scenes over the University of Houston campus and adjacent urban areas, consisted of HSI- and LiDAR-based DSM, both with a data size of 349×1905 pixels and a spatial resolution of 2.5 m. The HSI consists of 144 bands with a wavelength range of 0.38–1.05 μm. Also, LiDAR data are a single band. The dataset has a total of 15 029 ground-truth samples, covering 15 land covers.
2) *Trento Dataset:* The Trento dataset [48], [49] collects scenes from rural areas in southern Trento, Italy and includes HSI- and LiDAR-based DSMs. The data dimensions are 166×600 pixels with a spatial resolution of 1 m. HSI is collected by the Eagle sensor, an airborne hyperspectral imaging system, and contains 63 bands with a wavelength range of 0.42–0.99 μm. The LiDAR DSM is generated using the first and last point cloud pulses obtained from the Optech ALTM 3100EA sensor. The dataset contains a total of 30 214 truth value samples and studies six distinguishable class labels.
3) *MUUFL Dataset:* The MUUFL dataset [50] obtained HSI and LiDAR (DSM) data from The University of Southern Mississippi, Gulfport Campus in Long Beach, MS, USA. The dimensions are 325×220 pixels. HSI data are collected by ITRES' CASI-1500 sensor. HSI has 64 available spectral channels ranging from 0.38 to 1.05 μm with a spatial resolution of 0.54×1.0 m. LiDAR data are captured by an ALTM sensor using a laser at a wavelength of 1064 nm. The spatial resolution is 0.60×0.78 m. The dataset contains a total of 53 687 truth samples, and 11 distinguishable class labels are studied.
4) *Augsburg Data:* The Augsburg dataset [48] collects scenes over the city of Augsburg, Germany, including: HSI and LiDAR-based DSMs. The data size is 332×485 pixels. HSI is acquired by HySpex sensor acquisition. It contains 180 bands with wavelengths ranging from 0.4 to 2.5 μm. The LiDAR DSM is collected by the DLR-3k system. To facilitate the multimodal fusion, both images were downsampled to a uniform resolution of 30 m. The dataset contains a total of 78 294 truth samples, and seven distinguishable class labels are studied.

B. Experimental Setup

The proposed model is implemented in the PyTorch framework. The experiments are performed on the Ubuntu 22.04 platform. We use an AdamW optimizer with a learning rate decay parameter of 0.9 to optimize the network. In the training phase, the batch size and the number of training epochs are set to 32 and 1000, respectively. Considering four datasets have different data scales and spatial resolutions, initial learning rates are set to $4e^{-4}$ (Houston2013 dataset), $4e^{-4}$ (Trento dataset), $2e^{-4}$ (MUUFL dataset), and $4e^{-4}$ (Augsburg dataset). The standard cross entropy is utilized as the loss function.

To objectively evaluate the classification performance, three quantitative metrics are employed: overall accuracy (OA), average accuracy (AA), and the Kappa coefficient. Table II presents the number of training and testing samples utilized in this article.

C. Performance Comparison

Tables III–VI display the objective classification performances for various methods. In this study, we chose several representative models in recent years, such as EndNet [51], Coupled CNNs [19], NNCNet [28], GLT-Net [46], SS-MAE [52], MS2CANet [53], S2E [54], HCT-Net [55], and MHST [38], were compared and analyzed. These architectures can be broadly categorized into two types: those based solely on deep learning CNNs and those integrating CNNs with Transformers. The results indicate that the model employing the CNNs–Transformer architecture generally achieves superior accuracy compared with the model relying exclusively on the deep learning CNNs architecture.

Compared with other methods, the proposed approach demonstrates superior performance. On the Houston 2013 dataset, IDNet achieves an OA value that was 0.55% higher than the second-best model. For the Trento dataset, IDNet attains an OA of 99.62%, surpassing the second-best model by 0.17%. On the MUUFL dataset, IDNet records an OA of 89.12%, outperforming the second-best model by 0.41%. Finally, on the Augsburg dataset, IDNet achieves an OA of 88.74%, exceeding the second-best model by 2.58%. This performance improvement is attributed to the incorporation of the IC Transformer and ED-MoE, which

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TABLE II
DATASET DESCRIPTION

HOUSTON 2013 DATASET				TRENTO DATASET				MUUFL DATASET				AUGSBURG DATASET			
No.	Class	Training	Test	No.	Class	Training	Test	No.	Class	Training	Test	No.	Class	Training	Test
1	Health Grass	20	1231	1	Apple trees	20	4014	1	Trees	60	23186	1	Forest	20	13487
2	Stressed Grass	20	1234	2	Buildings	20	2883	2	Mostly grass	60	4210	2	Residential Area	20	30309
3	Synthetic Grass	20	677	3	Ground	20	459	3	Mixed ground surface	60	6822	3	Industrial Area	20	3831
4	Trees	20	1224	4	Woods	20	9103	4	Dirt and sand	60	1766	4	Low Plants	20	26837
5	Soil	20	1222	5	Vineyard	20	10481	5	Road	60	6627	5	Allotment	20	555
6	Water	20	305	6	Roads	20	3154	6	Water	60	406	6	Commercial Area	20	1625
7	Residential	20	1248					7	Buildings shadow	60	2173	7	Water	20	1510
8	Commercial	20	1224					8	Buildings	60	6180				
9	Road	20	1232					9	Sidewalk	60	1325				
10	Highway	20	1207					10	Yellow curb	60	123				
11	Railway	20	1215					11	Cloth panels	60	209				
12	Parking Lot 1	20	1213												
13	Parking Lot 2	20	449												
14	Tennis Court	20	408												
15	Running Track	20	640												

TABLE III
COMPARISON OF THE CLASSIFICATION ACCURACIES USING THE HOUSTON 2013 DATASET

No.	Class	EndNet	Coupled CNN	NNCNet	GLT-Net	SS-MAE	MS2CANet	S2E	HCT-Net	MHST	IDNet
1	Health Grass	91.36	94.24	97.80	94.09	86.51	**98.86**	97.40	89.03	98.05	98.05
2	Stressed Grass	86.47	97.21	97.40	96.74	**99.91**	84.84	98.78	96.52	98.22	93.19
3	Synthetic Grass	96.75	99.65	99.85	99.57	**100**	97.78	98.38	99.85	99.26	99.55
4	Trees	88.69	98.53	98.28	98.15	99.01	95.91	**100**	99.51	99.59	99.42
5	Soil	91.19	99.56	**100**	99.80	**100**	99.91	99.92	99.92	99.35	99.91
6	Water	93.15	98.23	**100**	97.08	**100**	**100**	**100**	**100**	99.67	**100**
7	Residential	59.07	94.88	93.34	94.94	96.62	94.95	96.31	**99.04**	95.59	95.43
8	Commercial	57.79	87.38	89.21	91.59	**91.74**	86.19	89.71	89.38	90.77	90.44
9	Road	66.94	85.43	78.89	**94.20**	83.82	84.98	89.04	84.01	89.20	87.90
10	Highway	69.15	94.69	95.77	91.92	97.10	96.68	91.21	92.87	96.77	**99.33**
11	Railway	64.05	95.80	98.51	93.18	95.46	90.45	94.73	94.16	94.32	**100**
12	Parking Lot 1	43.31	85.56	96.61	90.68	75.76	89.94	93.08	87.63	93.82	**98.68**
13	Parking Lot 2	43.92	98.60	94.20	98.40	99.77	97.32	95.32	**100**	97.33	96.88
14	Tennis Court	97.55	**100**	**100**	99.90	**100**	**100**	**100**	**100**	**100**	**100**
15	Running Track	98.61	99.69	**100**	99.98	99.84	98.28	**100**	**100**	**100**	**100**
	OA	74.44	94.35	95.30	95.32	93.83	93.31	95.65	94.02	96.19	**96.74**
	AA	76.08	95.30	95.99	96.01	95.04	94.40	96.26	95.19	96.80	**97.25**
	Kappa	72.87	93.89	94.91	94.95	93.33	92.76	95.29	93.53	95.88	**96.47**

TABLE IV
COMPARISON OF THE CLASSIFICATION ACCURACIES USING THE TRENTO DATASET

No.	Class	EndNet	Coupled CNN	NNCNet	GLT-Net	SS-MAE	MS2CANet	S2E	HCT-Net	MHST	IDNet
1	Apple trees	89.01	99.65	98.75	99.34	99.17	99.42	99.58	98.28	99.60	**99.77**
2	Buildings	92.04	96.60	94.58	97.68	97.18	97.08	96.43	94.48	98.06	**98.40**
3	Ground	86.41	95.75	99.34	97.89	99.78	97.82	97.82	**100**	99.78	99.78
4	Woods	91.09	99.59	99.67	99.93	99.98	99.79	**100**	**100**	**100**	**100**
5	Vineyard	81.94	98.96	95.85	99.80	99.45	99.93	**99.99**	99.55	**99.99**	99.96
6	Roads	86.48	95.72	94.26	97.17	97.71	86.55	98.10	92.61	97.08	**98.31**
	OA	87.16	98.62	97.16	99.27	99.18	98.11	99.37	98.31	99.45	**99.62**
	AA	87.83	97.71	97.08	98.63	98.88	96.76	98.65	97.49	99.09	**99.37**
	Kappa	83.21	98.17	96.22	99.03	98.19	97.48	99.15	97.75	99.26	**99.49**

promote shared feature representations across domains, thereby enhancing the multimodal feature fusion.

1) Quantitative Index Analysis: Tables III–VI present the quantitative classification results of various methods on four widely used HSI and LiDAR datasets, respectively. Due to the scarcity of a small number of samples, the category imbalance makes it difficult for the model to capture its statistical patterns. At the same time, our model contains deep convolution and channel attention mechanisms, but the amount of data in the MUUFL dataset is insufficient, for example, there are only 223 samples of yellow curb, 309 samples of cloth panels, and 506 samples of water, resulting in information loss and overfitting, and when dealing with complex image classification, there are fewer deep convolutional parameters, resulting in a slightly lower AA. At the same time, assigning the weight of the channel attention module on a small dataset can easily lead to the reduction of AA. A comprehensive analysis reveals that IDNet consistently delivers outstanding performance, achieving the highest classification scores across all four datasets. By thoroughly examining the characteristics of land cover categories and their corresponding classification accuracies, we derive the following conclusions.

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TABLE V
COMPARISON OF THE CLASSIFICATION ACCURACIES USING THE MUUFL DATASET

No.	Class	EndNet	Coupled CNN	NNCNet	GLT-Net	SS-MAE	MS2CANet	S2E	HCT-Net	MHST	IDNet
1	Trees	82.04	89.12	84.95	92.20	92.59	78.44	89.64	87.32	91.81	**93.24**
2	Mostly grass	79.48	73.93	83.30	80.59	79.36	77.03	81.00	75.99	**85.70**	81.40
3	Mixed ground surface	64.00	58.47	75.47	70.57	76.87	69.83	67.40	69.29	72.78	**79.17**
4	Dirt and sand	89.47	86.36	88.10	80.41	83.74	**97.17**	94.73	95.13	89.81	93.82
5	Road	86.10	80.46	79.34	**88.77**	87.01	41.41	81.80	81.67	88.29	82.78
6	Water	98.77	99.53	99.26	**100**	**100**	**100**	**100**	**100**	**100**	99.75
7	Buildings shadow	89.78	82.42	87.89	**96.36**	95.16	78.58	94.43	90.29	93.05	90.01
8	Buildings	87.61	95.39	94.11	93.41	90.72	84.88	92.51	94.66	95.52	**97.55**
9	Sidewalk	73.28	63.32	58.56	**84.68**	83.01	29.59	76.08	64.60	82.19	72.22
10	Yellow curb	**95.12**	82.15	91.86	79.67	86.06	87.73	92.68	82.93	92.68	93.59
11	Cloth panels	98.09	95.14	95.21	99.52	99.04	**99.59**	99.52	99.52	99.04	99.52
	OA	81.24	82.69	83.70	87.86	88.24	73.05	85.60	84.20	88.71	**89.12**
	AA	85.79	82.39	85.28	87.84	88.51	76.75	88.16	85.58	**90.08**	89.36
	Kappa	76.08	77.70	79.06	84.20	84.66	66.23	81.42	79.67	85.32	**85.78**

TABLE VI
COMPARISON OF THE CLASSIFICATION ACCURACIES USING THE AUGSBURG DATASET

No.	Class	EndNet	Coupled CNN	NNCNet	GLT-Net	SS-MAE	MS2CANet	S2E	HCT-Net	MHST	IDNet
1	Forest	92.91	97.57	96.82	96.30	98.64	96.63	98.43	95.37	**99.20**	97.75
2	Residential Area	32.68	75.47	78.54	77.45	78.26	71.36	65.58	65.87	82.88	**89.22**
3	Industrial Area	48.26	81.83	58.23	69.04	68.91	66.80	53.09	39.08	66.35	**82.27**
4	Low Plants	66.92	87.46	**91.92**	84.06	85.68	54.43	80.04	87.03	89.70	88.64
5	Allotment	68.47	93.69	91.71	90.81	**97.29**	93.69	96.22	94.95	92.25	81.08
6	Commercial Area	30.22	32.80	55.50	50.95	45.78	43.26	51.45	**63.38**	50.64	47.38
7	Water	**73.18**	15.10	70.19	51.05	66.68	72.91	69.87	62.05	58.54	63.90
	OA	56.58	82.80	84.75	81.59	83.10	69.29	75.61	76.99	86.16	**88.74**
	AA	58.95	76.61	77.56	74.24	77.32	71.30	73.52	72.53	77.08	**78.61**
	Kappa	45.91	76.70	79.15	74.98	76.84	59.87	67.76	69.58	80.93	**84.33**

While extracting features from the HSI data with numerous spectral channels, the S^2IC module demonstrates superior effectiveness in comprehensively capturing spectral features compared with the conventional Transformer. Within the HSI learning branch, we specifically design a novel S^2IC to optimize spectral information extraction from HSI. For example, accurately classifying land cover categories such as "highway" and "railway" poses significant challenges on the Houston 2013 dataset, while relying solely on traditional single-modal data. Compared with methods, i.e., GLT-Net and MHST, our approach achieves an improvement of over 2% in classification accuracy for these land covers.

In the feature fusion stage, the differential convolution is utilized to enhance the detail experts, with the objective of further improving the classification accuracy. Unlike other methods that depend solely on feature-level fusion, our approach emphasizes the significance of high-frequency feature information, thereby enhancing the resolution of the fused features. For example, distinguishing between the "residential area" and "industrial area" categories presents a considerable challenge on the Augsburg dataset; however, our method surpasses HCT-Net and S2E by more than 10%.

2) Visual Comparative Analysis: Figs. 6–9 illustrate the classification maps generated by various methods across four different datasets. It is evident that the visual classification results produced by IDNet exhibit a closer alignment with the ground-truth maps compared with those generated by other approaches.

Specifically, EndNet exhibits a substantial number of classification errors on the Trento dataset, with numerous isolated data points in NNCNet, GLT-Net, and S2E. On the MUUFL dataset, IDNet's results are more closely aligned with the ground truth, with clearer boundaries compared with other methods. Furthermore, both GLT-Net and HCT-Net misclassify "residential area" as "water," whereas IDNet's classification results on the Augsburg dataset are nearly flawless. Toward the end, IDNet demonstrates the superior classification performance.

3) Computational Complexity Analysis: Table VII presents the computational complexity of different methods, including the trainable parameters and giga floating-point operations per second (GFLOPs), a metric quantifying the number of billion floating-point calculations per second) on the Augsburg dataset. It can be seen that the coupled CNN method can train with the fewest parameters, while the NNCNet method uses the most parameters. The parameter of this model is less than that of other networks, and GFLOPs are larger. Within an acceptable number of trainable parameters and GFLOPs, IDNet exhibits the best classification performance.

D. Parameters Analysis

1) Patch Size: The classification performance of a model is notably influenced by the varying sizes of the input patches. Based on the network architecture, we select patch sizes of 8×8, 12×12, and 16×16 to conduct experimental evaluations across four different datasets. To quantitatively evaluate the performance of our model, we employ OA as the primary evaluation metric.

The experimental results are presented in Fig. 10. Specifically, for the Trento and Houston 2013 datasets, our model

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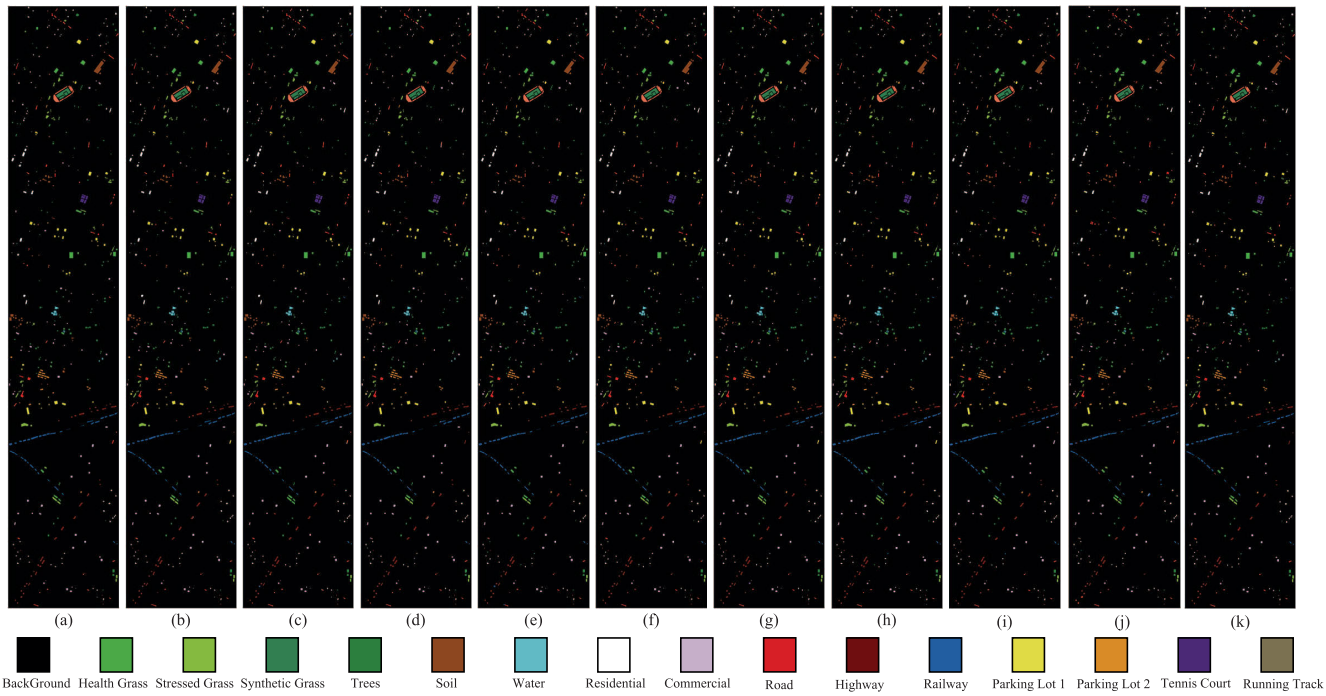


Fig. 6. Classification maps by different methods on the Houston2013 dataset. (a) Ground-truth map. (b) EndNet (74.44%). (c) Coupled CNN (94.35%). (d) NNCNet (95.30%). (e) GLT-Net (95.32%). (f) SS-MAE (93.83%). (g) MS2CANet (93.31%). (h) S2E (95.65%). (i) HCT-Net (94.02%). (j) MHST (96.19%). (k) IDNet (96.74%).

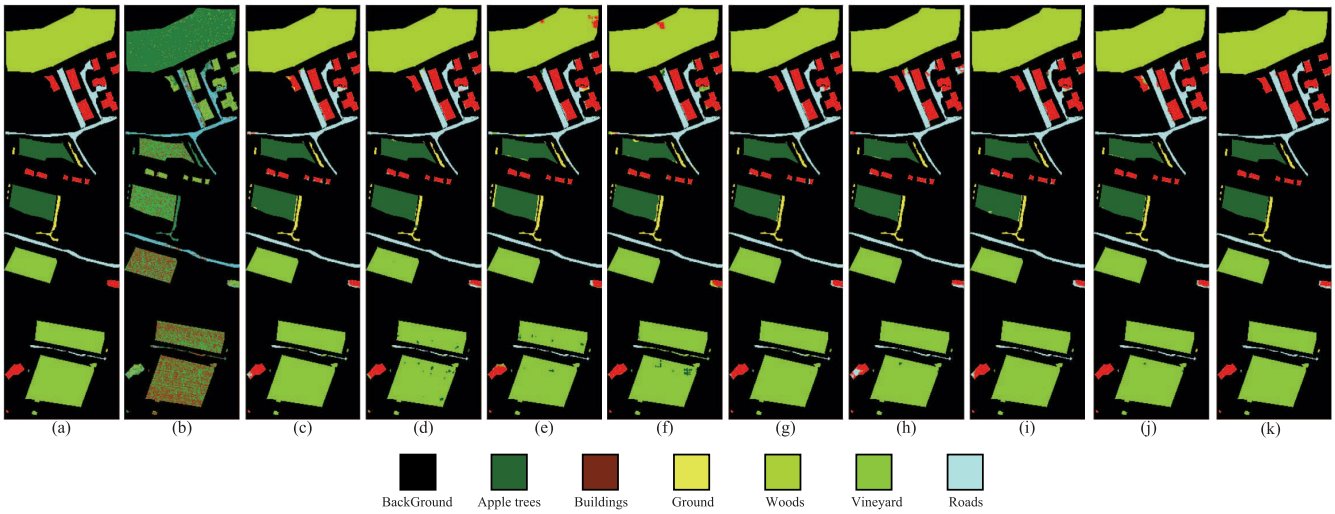


Fig. 7. Classification maps by different methods on the Trento dataset. (a) Ground-truth map. (b) EndNet (87.16%). (c) Coupled CNN (98.62%). (d) NNCNet (97.16%). (e) GLT-Net (99.27%). (f) SS-MAE (99.18%). (g) MS2CANet (98.11%). (h) S2E (99.37%). (i) HCT-Net (98.31%). (j) MHST (99.45%). (k) IDNet (99.62%).

TABLE VII

COMPARISON OF COMPUTATIONAL COMPLEXITY AND TESTING TIME (S) OF DIFFERENT METHODS ON THE AUGSBURG DATA

Methods	EndNet	Coupled CNN	NNCNet	GLT-Net	SS-MAE	MS2CANet	S2E	HCT-Net	MHST	IDNet
Trainable Params	139k	104k	2M	3M	2M	696k	139k	429k	336k	111k
GFLOPs	2.82×10^{-3}	169.11×10^{-3}	9.74	7.97×10^{-3}	8.53×10^{-3}	8.72×10^{-3}	6.83×10^{-3}	7.93×10^{-3}	1.81	1.53

achieves optimal performance with a patch size of 8×8. As the patch size increases, a significant reduction in classification accuracy is observed. This indicates that, for these two datasets, smaller patch sizes are more effective in capturing the essential features and nuances required for accurate classification.

Conversely, for the MUUFL and Augsburg datasets, our model yields optimal results with a patch size of 12×12.

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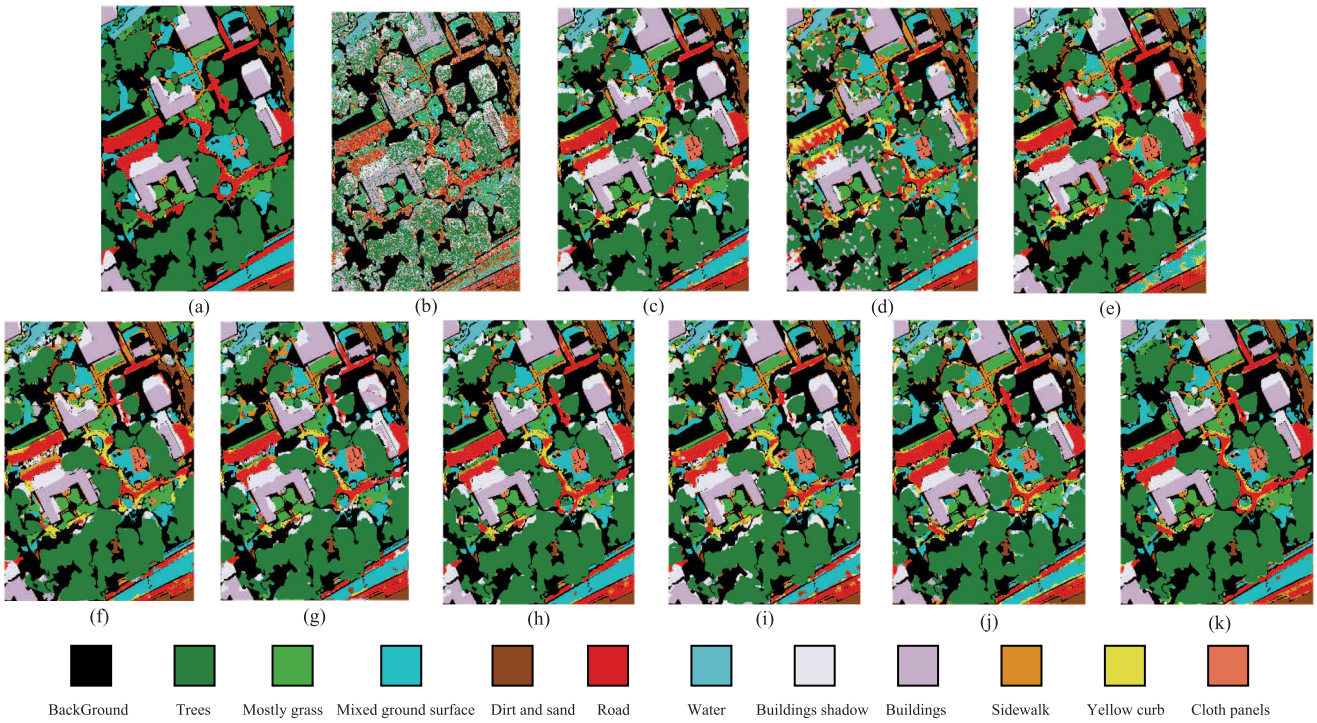


Fig. 8. Classification maps by different methods on MUUFL dataset. (a) Ground-truth map. (b) EndNet (81.24%). (c) Coupled CNN (82.69%). (d) NNCNet (83.70%). (e) GLT-Net (87.86%). (f) SS-MAE (88.24%). (g) MS2CANet (73.05%). (h) S2E (85.60%). (i) HCT-Net (84.20%). (j) MHST (88.71%). (k) IDNet (89.12%).

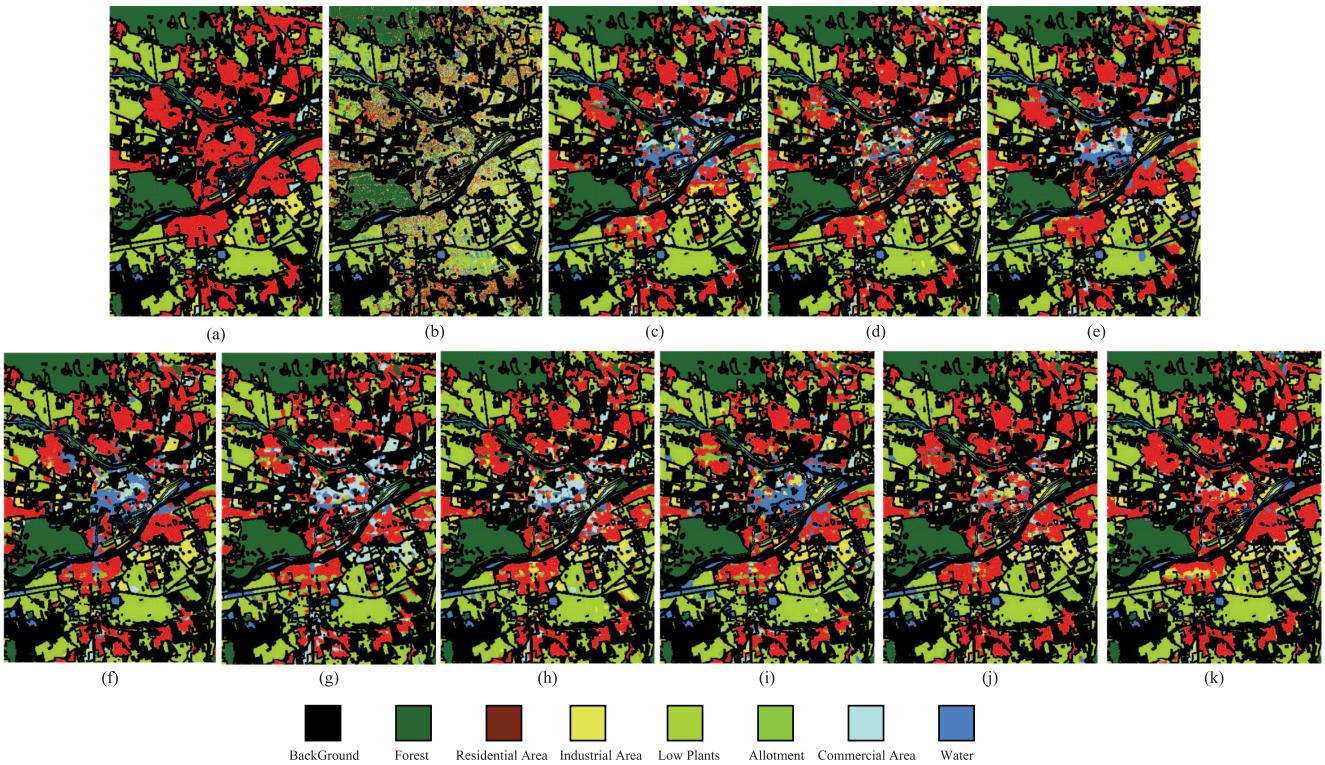


Fig. 9. Classification maps by different methods on the Augsburg dataset. (a) Ground-truth map. (b) EndNet (56.58%). (c) Coupled CNN (82.80%). (d) NNCNet (84.75%). (e) GLT-Net (81.59%). (f) SS-MAE (83.10%). (g) MS2CANet (69.29%). (h) S2E (75.61%). (i) HCT-Net (76.99%). (j) MHST (86.16%). (k) IDNet (88.74%).

This suggests that, for these datasets, a larger patch size is more effective in capturing the relevant information and spatial relationships critical for achieving high classification accuracy. These findings emphasize the significance of carefully

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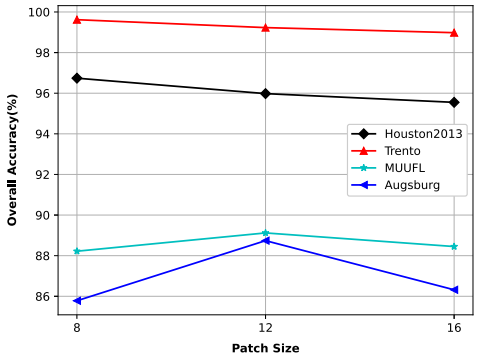


Fig. 10. Effects of the patch on classification performance.

selecting the patch size in accordance with the characteristics of the dataset, as the selected patch size could significantly influence the model's performance.

2) HSI Bands: We employ PCA for the dimensionality reduction of hyperspectral imagery, with the objective of reducing model complexity, preserving the essential information, and mitigating overfitting. However, it is crucial to select an appropriate number of channels that align with the model's architecture and the characteristics of the dataset. Therefore, the determination of the number of hyperspectral channels is a critical parameter that significantly impacts the model's classification performance. To investigate this, we conduct an experiment to evaluate the effect of varying the number of HSI channels, setting the number of bands to 10, 20, and 30. The evaluation metrics used are OA, AA, and Kappa.

The HSI bands are pivotal in determining the model's performance, which prompts a thorough investigation into the effects of varying this parameter. We systematically conduct an analysis, starting with ten HSI bands and progressively increasing the number to 30, while consistently employing OA, AA, and Kappa as the primary metrics to evaluate the model's performance. As illustrated in Fig. 11, the optimal results are achieved with 30 HSI channels for the Houston2013, Trento, and Augsburg datasets, whereas the highest classification accuracy is attained with 20 channels on the MUUFL dataset.

3) Experts Network Number: The number of expert networks within a model can significantly impact its classification performance. To explore this, we conduct a series of experiments, beginning with a baseline of two expert networks and progressively increasing the count by one in the following experiment. Then, we utilize OA, AA, and Kappa as the primary metrics.

The experimental results are illustrated in Fig. 12. The optimal classification performance is achieved when the number of expert networks is set to 10. This configuration enables the model to harness the advantages of multiple expert networks without introducing unnecessary complexity or redundancy. In contrast, any deviation from this optimal number, whether an increase or decrease, could negatively impact classification performance. Specifically, increasing the number of expert networks beyond 10 led to overfitting and heightened computational complexity, whereas reducing the number resulted in underfitting and insufficient diversity in the model's decision-making stage. Both classification performances ultimately caused a decline in the model's accuracy, highlighting the critical importance of selecting an appropriate number of expert networks to achieve the optimal performance.

4) Weight Coefficient: In the complex feature fusion between HSI and LiDAR data, the weight coefficient plays a pivotal role in determining the classification performance. We conducted a series of experiments, starting from a weight coefficient of 0.1. In the following experiments, this coefficient is incrementally adjusted by 0.1, up to a maximum value of 0.9, with the OA being closely monitored as the primary metric for evaluation.

The results, as shown in Fig. 13, demonstrate that while the weight coefficient is precisely set to 0.5, IDNet achieves optimal classification performance across all four datasets. This configuration allows the model to fully leverage the complementary strengths of HSI and LiDAR fusion features, resulting in the most accurate and detailed classification performances. In contrast, even a slight deviation from this optimal coefficient, whether through an increase or decrease, leads to a substantial decline. This result underscores the delicate balance required in integrating HSI and LiDAR fusion features and emphasizes the critical importance of fine-tuning the weight coefficients to ensure the optimal spectral feature extraction and classification accuracy.

E. Ablation Studies

This section presents ablation experiments designed to evaluate the efficacy of various components. The corresponding experimental results are illustrated in Tables VIII.

1) Effectiveness of the Spatial Intensity-Constrained Block: To evaluate the contribution of the spatial intensity-constrained block, we devised a modified version of our proposed model, designated as variant-w/o spatial intensity-constrained block. This variant omits the spatial intensity-constrained block, thereby facilitating a direct comparison of its impact on model performance. IDNet, incorporating the spatial intensity-constrained block, demonstrates superior performance across all datasets. Specifically, it achieves improvements in OA by 1.35%, 0.53%, 2.14%, and 0.27%, respectively, while compared to the variant-w/o spatial intensity-constrained block. These enhancements not only substantiate the efficacy of the spatial intensity-constrained block but also highlight its pivotal role in augmenting the model's capacity to accurately classify and interpret data. By integrating the spatial intensity-constrained block, IDNet is better equipped to exploit the intricate patterns and features present in the hyperspectral and LiDAR data, resulting in more robust and reliable classification performances.

2) Effectiveness of the Spatial–Spectral Intensity-Constrained Block: To verify the effectiveness and significance of the spatial–spectral intensity-constrained block, we developed a variant of our proposed model, referred to as variant-w/o spatial–spectral intensity-constrained block. This variant omits the spatial–spectral intensity-constrained block, facilitating a direct comparison to evaluate its impact on model performance. Extensive experiments were conducted across four diverse datasets to compare the performance of IDNet,

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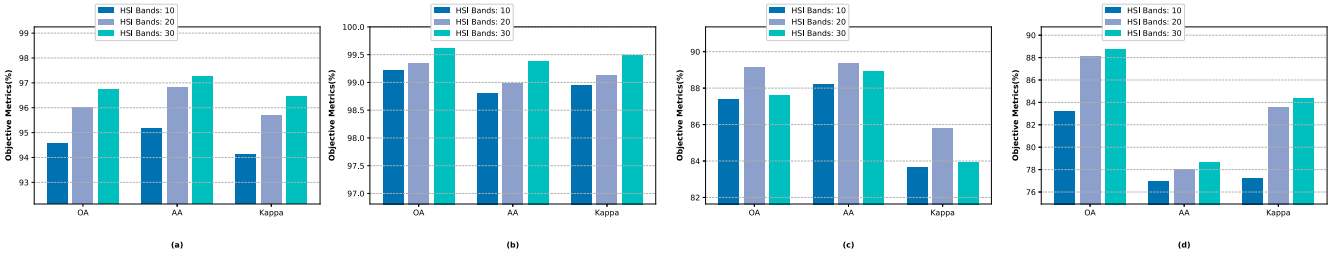


Fig. 11. Effects of the HSI bands on classification performance. (a) Houston2013. (b) Trento. (c) MUUFL. (d) Augsburg.

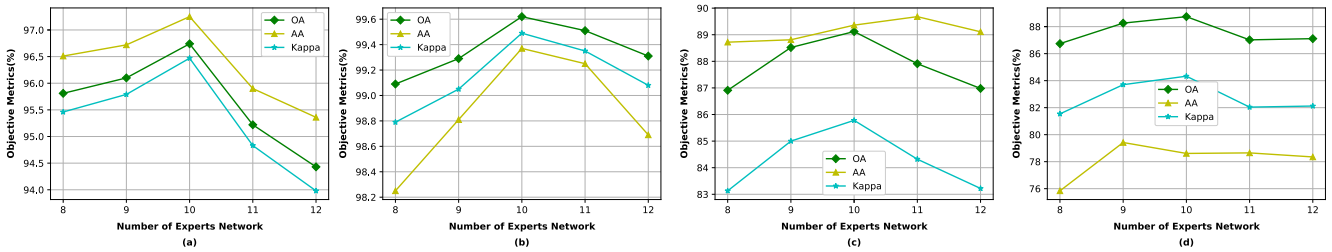


Fig. 12. Effects of the experts' network number on classification performance. (a) Houston2013. (b) Trento. (c) MUUFL. (d) Augsburg.

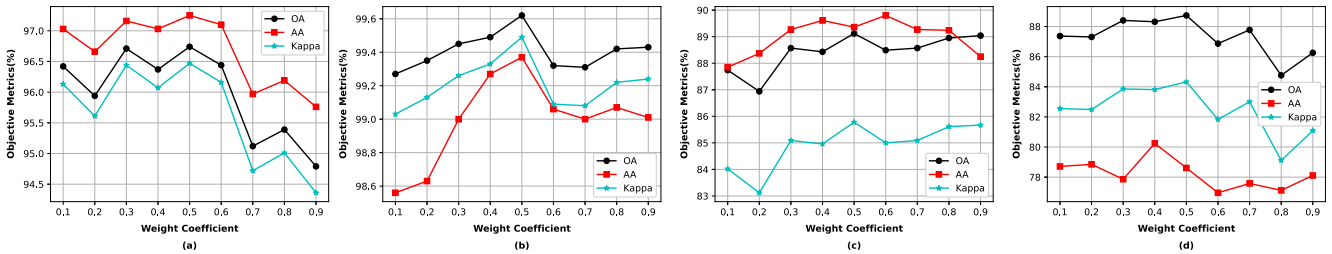


Fig. 13. Effects of the weight parameter on classification performance. (a) Houston2013. (b) Trento. (c) MUUFL. (d) Augsburg.

TABLE VIII

INFLUENCE OF SPATIAL INTENSITY-CONSTRAINED BLOCK, SPATIAL–SPECTRAL INTENSITY-CONSTRAINED BLOCK, AND ED-MOE ON CLASSIFICATION RESULTS OF IDNET

SIC block	S^2IC block	EDME	Laplacian	Houston2013	Trento	MUUFL	Augsburg
✗	✔	✔	✔	95.39	99.09	86.98	88.47
✔	✗	✔	✔	94.81	99.13	86.91	85.90
✔	✔	✗	✔	94.99	99.39	88.77	86.26
✔	✔	✔	✗	93.26	98.83	86.26	85.74
✔	✔	✔	✔	96.74	99.62	89.12	88.74

which incorporates the spatial–spectral intensity-constrained block, with the variant lacking this component. The results were both clear and consistent: IDNet outperformed the variant in all datasets. Specifically, it achieves improvements in OA of 1.93%, 0.49%, 2.21%, and 2.84%, respectively, relative to the variant without the spatial–spectral intensity-constrained block. These enhancements not only substantiate the efficacy of the spatial–spectral intensity-constrained block but also emphasize its critical role in enhancing the model's ability to accurately capture and interpret both spatial and spectral information within the data. By incorporating the spatial–spectral intensity-constrained block, IDNet is better positioned to exploit the complex patterns and features inherent in hyperspectral and LiDAR data, resulting in more robust and reliable classification performances. These results further advocate for the adoption of the spatial–spectral intensity-constrained block in similar applications, where precise spatial and spectral information are vital for effective data analysis and interpretation.

3) Effectiveness of the ED-MoE: To demonstrate the effectiveness and superiority of the ED-MoE approach, we develop a variant of our proposed model, termed variant-w/o ED-MoE. This variant intentionally omits the EDMoE component, facilitating a controlled comparison to evaluate its impact on model performance. The evaluations are conducted across four datasets to evaluate the performance differences between IDNet, which incorporates the EDMoE, and the variant-w/o ED-MoE. Specifically, IDNet achieves improvements in OA of 1.75%, 0.23%, and 0.35%, and a notable 2.48%, respectively, compared with the method without the EDMoE. These

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enhancements not only validate the efficacy of the EDMoE but also highlight its crucial role in enhancing the model's ability to capture intricate details within the data. By integrating the EDMoE, IDNet is better positioned to leverage the diverse and complex information embedded in the datasets, resulting in more accurate and reliable classification performances.

4) Effectiveness of the Laplacian: In order to prove the validity and superiority of the Laplace algorithm, we developed a variant of our proposed model called variant-w/O Laplace. This variant deliberately omits the Laplace component to facilitate a controlled comparison to assess its impact on model performance. The evaluation was conducted on four datasets to assess the performance differences between IDNets. Compared with the method without Laplace, the OA performance of IDNet was improved by 3.48%, 0.79%, 2.86%, and 3.00%, respectively. These enhancements not only validate Laplace's effectiveness but also underscore its critical role in obtaining high-frequency information. By integrating Laplace, IDNet can better leverage the diverse and complex information embedded in the dataset for more accurate and reliable classification performance.

V. Conclusion

In this article, we start from the high-frequency detail features of HSI and LiDAR data, focusing on spatial–spectral high-frequency detail feature learning and propose an IDNet to enhance the feature extraction and classification capabilities of HSI and LiDAR data under a complex background. Specifically, we introduce spatial–spectral intensity-constrained blocks in the hyperspectral feature learning branch and spatial intensity-constrained blocks in the LiDAR feature learning branch. These blocks exploit Laplace high-frequency features for dynamic range perception and facilitate feature grouping interactions, thereby enabling the effective extraction of local high-frequency features. Furthermore, during the feature fusion stage, we incorporate the enhanced detail mixture experts module, which employs differential convolution and a dynamic selection mechanism to emphasize high-frequency feature information, enhancing the detailed feature description and discriminability of the fused features. Verification experiments on four challenging datasets demonstrate the superior performance of the proposed network.

Although the proposed network has validated the effectiveness of high-frequency feature learning in the collaborative classification of HSI and LiDAR, low-frequency features constitute the majority of the image information, and the general outline and contours of the image provide approximate information. The effectiveness of these features in HSI and LiDAR classification remains to be verified. Furthermore, the selection and fusion of high-frequency and low-frequency features will be a key focus in future research.

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